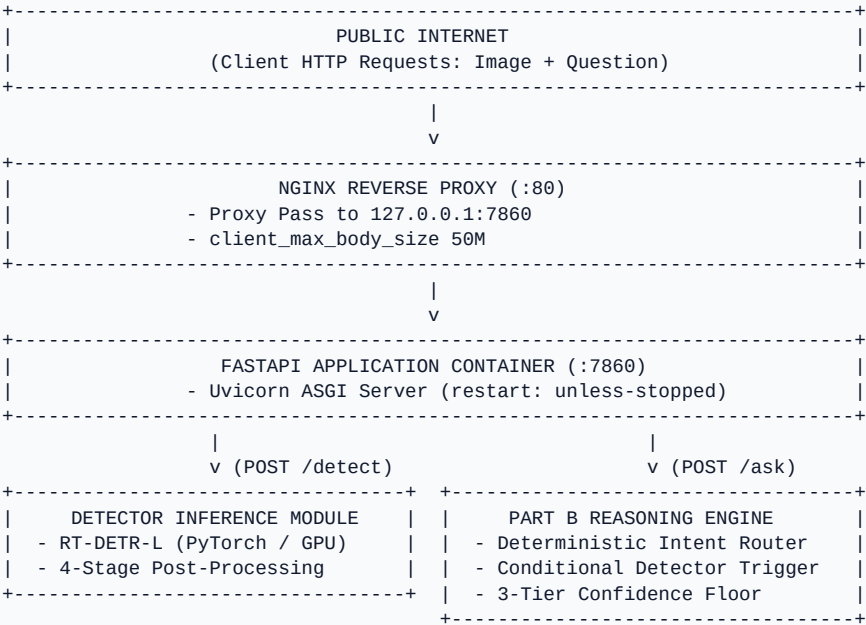


System Design & Architecture Specification

END-TO-END CV + REASONING PIPELINE ARCHITECTURE

1. High-Level Architecture Diagram



2. Core Subsystem Responsibilities

- **Dataset Pipeline:** 2,500 image benchmark dataset (1500 train, 500 val, 500 test).
- **Model Inference Engine:** RT-DETR-L Transformer with 4-stage post-processing (Class-aware NMS, Same-class IoS containment, Cross-class IoU, Boundary filter).
- **Reasoning Layer:** Deterministic intent router handling COUNT, PRESENCE, LIST, SPATIAL, MOST_COMMON, and UNKNOWN queries.
- **Guardrail Enforcement:** Prevents hallucination by truncating low-confidence detections (<0.25).
- **Infrastructure:** Managed via Terraform IaC on AWS EC2 t3.small with Docker Compose and Nginx.